Cancer is **both genetic and caused by other factors**.

At its core, cancer happens because of changes or mutations in DNA. So in that sense, cancer is always a genetic disease. But those mutations are often caused by things other than the genes you inherited.

Most cancers are linked to a mix of:

- Aging and random mistakes when cells divide
- Smoking
- Alcohol
- Sun exposure / UV radiation
- Pollution or certain chemicals
- Viruses like Human papillomavirus infection
- Diet and lifestyle
- Family history / inherited genes

Only about 5–10% of cancers are mainly due to inherited mutations passed down from parents, such as changes in the BRCA1 or BRCA2 genes. The rest usually develop over time because of environmental exposures, lifestyle, and chance.

So a simple way to think about it is:

- **Inherited genes** can raise your risk
- **Lifestyle and environment** can trigger mutations
- **Random cell mistakes** also play a role over time

All three can contribute together.

gene and protein in disease impact factor:

Proteins are the entities that actually interact with each other inside the body.

A gene is mainly an instruction set stored in DNA.

That gene produces a protein.

Then proteins:

- Bind to each other
- Activate each other
- Inhibit each other
- Work together in pathways
- Form complexes
- Pass signals

Example:

- BRCA1 protein helps repair DNA
- BRCA2 protein also helps repair DNA
- Their proteins interact in the same repair mechanism

So databases like STRING store these protein-protein interactions.

1. NETWORK BIOLOGY: UNDERSTANDING THE CELL'S FUNCTIONAL ORGANIZATION
- Albert-László Barabási* & Zoltán N. Oltvai‡

- most biological characteristics arise from complex interactions between the cell's numerous constituents, such as proteins, DNA, RNA and small molecules1–8. ,
  Inference: need to study how each individual molecule works in the group
  a key challenge for biology in the twenty-first century is to understand the structure and the dynamics of the complex intercellular web of interactions that contribute to the structure and function of a living cell.

, new technology platforms, such as PROTEIN CHIPS or semi-automated YEAST TWO-HYBRID SCREENS, help to determine how and when these molecules interact with each other. Various types of interaction webs, or networks, (including protein–protein interaction, metabolic, signalling and transcription-regulatory networks) emerge from the sum of these interactions. None of these networks are independent, instead they form a 'network of networks' that is responsible for the behaviour of the cell.

similar laws may govern most complex networks in nature, which allows the expertise from large and well-mapped non-biological systems to be used to characterize the intricate interwoven relationships that govern cellular functions

most basic network measures that allow us to compare and characterize different complex networks:

Degree The most elementary characteristic of a node is its degree (or connectivity), k, which tells us how many links the node has to other nodes.

Degree distribution The degree distribution,P(k), gives the probability that a selected node has exactly k links.P(k) is obtained by counting the number o f nodes N(k) with k = 1, 2… links and dividing by the total number of nodes N. The degree distribution allows us to distinguish between different classes of networks. For example, a peaked degree distribution, as seen in a random network (BOX 2), indicates that the system has a characteristic degree and that there are no highly connected nodes (which are also known as hubs). By contrast, a power-law degree distribution indicates that a few hubs hold together numerous small nodes (BOX 2).

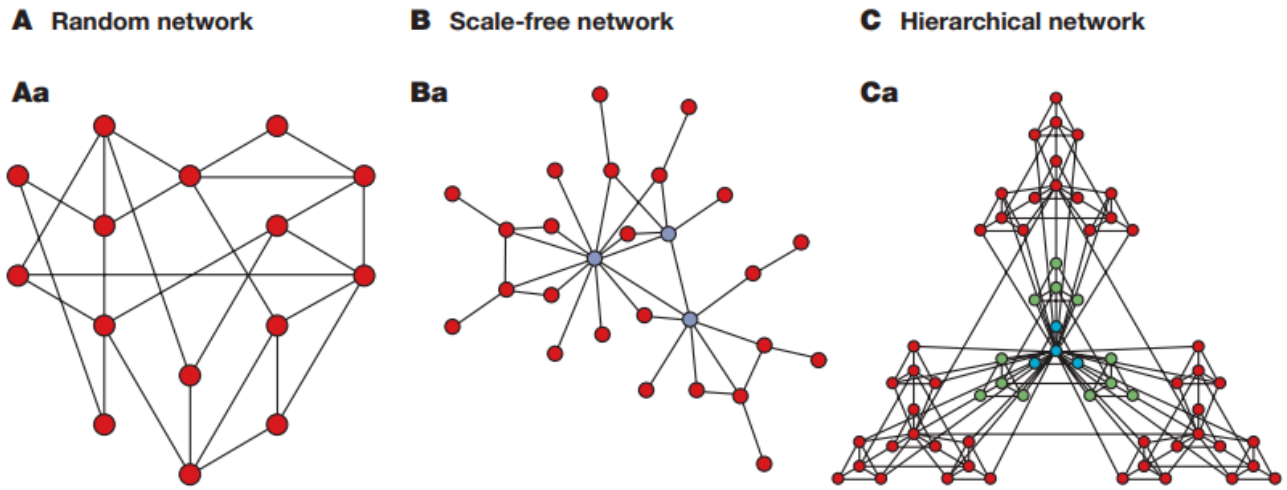
Scale-free networks and the degree exponent Most biological networks are scale-free, which means that their degree distribution approximates a power law,$P(k) \sim k^{-\gamma}$, where γ is the degree exponent and ~ indicates 'proportional to'. The value of γ determines many properties of the system. The smaller the value of γ, the more important the role of the hubs is in the network

Shortest path and mean path length Distance in networks is measured with the path length, which tells us how many links we need to pass through to travel between two nodes. The mean path length, < >, represents the average over the shortest paths between all pairs of nodes and offers a measure of a network's overall navigability.

Clustering coefficient In many networks, if node A is connected to B, and B is connected to C, then it is highly probable that A also has a direct link to C. This phenomenon can be quantified using the clustering coefficient33 $C_I = 2n_I / k(k-1)$, where n_I is the number of links connecting the k_I neighbours of node I to each other. The average degree , average path length < > and average

clustering coefficient depend on the number of nodes and links (N and L) in the network. By contrast, the P(k) and C(k) functions are independent of the network's size and they therefore capture a network's generic features, which allows them to be used to classify various networks.

- Network Models
  are three models that had a direct impact on our understanding of biological networks
  1. Random networks
     starts with N nodes and connects each pair of nodes with probability p, which creates a graph with approximately pN(N–1)/2 randomly placed links (see figure, part Aa). The node degrees follow a Poisson distribution (see figure, part Ab), which indicates that most nodes have approximately the same number of links (close to the average degree). The tail (high k region) of the degree distribution P(k) decreases exponentially, which indicates that nodes that significantly deviate from the average are extremely rare. The clustering coefficient is independent of a node's degree, so C(k) appears as a horizontal line if plotted as a function of k (see figure, part Ac). The mean path length is proportional to the logarithm of the network size, l ~ log N, which indicates that it is characterized by the small-world property.
  2. Scale-free networks
     Scale-free networks (see figure, part B) are characterized by a power-law degree distribution; the probability that a node has k links follows P(k) ~ k –γ , where γ is the degree exponent. The probability that a node is highly connected is statistically more significant than in a random graph, the network's properties often being determined by a relatively small number of highly connected nodes that are known as hubs (see figure, part Ba; blue nodes). In the Barabási–Albert model of a scale-free network15, at each time point a node with M links is added to the network, which connects to an already existing node I with probability ΠI = kI /ΣJ kJ , where kI is the degree of node I (FIG. 3) and J is the index denoting the sum over network nodes. The network that is generated by this growth process has a power-law degree distribution that is characterized by the degree exponent γ = 3. Such distributions are seen as a straight line on a log–log plot (see figure, part Bb). The network that is created by the Barabási–Albert model does not have an inherent modularity, so C(k) is independent of k (see figure, part Bc). Scale-free networks with degree exponents 2<γ<3, a range that is observed in most biological and non-biological networks, are ultra-small34,35, with the average path length following ~ log log N, which is significantly shorter than log N that characterizes random small-world networks.
  3. Hierarchical networks
     To account for the coexistence of modularity, local clustering and scale-free topology in many real systems it has to be assumed that clusters combine in an iterative manner, generating a hierarchical network uces a new module of 64 nodes. The hierarchical network model seamlessly integrates a scale-free topology with an inherent modular structure by generating a network that has a power-law degree distribution with degree exponent γ = 1 + n4/n3 = 2.26 (see figure, part Cb) and a large, system-size independent average clustering coefficient ~ 0.6. The most important signature of hierarchical modularity is the scaling of the clustering coefficient, which follows C(k) ~ k –1 a straight line of slope –1 on a log–log plot (see figure, part Cc). A hierarchical architecture implies that sparsely connected nodes are part of highly clustered areas, with communication between the different highly clustered neighbourhoods being maintained by a few hubs



A key feature of many complex systems is their robustness, which refers to the system's ability to respond to changes in the external conditions or internal organization while maintaining relatively normal behaviour. To understand the cell's functional organization, insights into the interplay between the network structure and robustness, as well as their joint evolutionary origins, are needed.

Topological robustness. Intuition tells us that disabling a substantial number of nodes will result in an inevitable functional disintegration of a network. This is certainly true for a random network: if a critical fraction of nodes is removed, a phase transition is observed, breaking the network into tiny, non-communicating islands of nodes.
Complex systems, from the cell to the Internet, can be amazingly resilient against component failure, withstanding even the incapacitation of many of their individual components and many changes in external conditions. We have recently learnt that topology has an important role in generating this topological robustness71. Scale-free networks do not have a critical threshold for disintegration — they are amazingly robust against accidental failures: even if 80% of randomly selected nodes fail, the remaining 20% still form a compact cluster with a path connecting any two nodes. This is because random failure affects mainly the numerous small degree nodes, the absence of which doesn't disrupt the network's integrity71. This reliance on hubs, on the other hand, induces a so-called attack vulnerability — the removal of a few key hubs splinters the system into small isolated node clusters71.

Functional and dynamical robustness. A complete understanding of network robustness requires that the functional and dynamic changes that are caused by perturbations are explored. In a cellular network, each node has a slightly different biological function and therefore the effect of a perturbation cannot depend on the node's degree only.

, it is increasingly accepted that adaptation and robustness are inherent network properties, and not a result of the fine-tuning of a component's characteristics80,81
Second, robustness is inevitably accompanied by vulnerabilities: although many cellular networks are well adapted to compensate for the most common perturbations, they collapse when wellselected network components are disrupted. Third, the ability of a module to evolve also has a key role in developing or limiting robustness8

2. Efficient Behavior of Small-World Networks
Vito Latora and Massimo Marchiori

any complex system in nature can be modeled as a network, where vertices are the elements of the system and edges represent the interactions between them.

Characterizing the structural properties of the networks is then of fundamental importance to understand the complex dynamics of these systems. A recent paper [2] has shown that the connection topology of some biological and social networks is neither completely regular nor completely random.

efficiency of a network, measuring how efficiently information is exchanged over the network. By using efficiency small-world networks results as systems that are both globally and locally efficient.

3. How to Understand the Cell by Breaking It: Network Analysis of Gene Perturbation Screens Florian Markowetz*

modern technologies make it possible to probe gene function on a genome-wide scale in many model organisms and human.
Observed phenotypes can reveal which genes are essential for an organism, or work in a particular pathway, or have a specific cellular function. Combining highthroughput screening techniques with rich phenotypes enables researchers to observe detailed reactions to experimental perturbations on a genome-wide scale.

Advances in the design and analysis of gene perturbation screens may have an immediate impact on many areas of biological and medical research.

Phenotypes: the set of observable characteristics of an individual resulting from the interaction of its genotype with the environment.
New screening and phenotyping techniques often directly translate into new insights in gene and protein functions.
Results of perturbation screens can also reveal unexploited areas of potential therapeutic intervention.
Novel ideas on how to use perturbation screens to uncover cellular wiring diagrams can lead to a better understanding of how cellular networks are deregulated in diseases like cancer. This knowledge is indispensable for finding new drug targets to attack the drivers of a disease and not only the symptoms.
. Analysis strategies strongly depend on how rich and informative phenotype descriptors are. We will call phenotypes resulting from a single reporter (or a small number of reporters) low-dimensional phenotypes and the genes showing significant results hits
Low-dimensional phenotyping screens can identify candidate genes on a genome-wide scale and are often used as a first step for follow-up analysis. We will discuss methods to functionally interpret hits from low-dimensional phenotyping screens and to place them in the context of cellular networks in the first part of this review. The second part will be devoted to high-dimensional phenotyping screens, which evaluate a large number of cellular features at the same time. Observing system-wide changes promises key insights into cellular mechanisms and pathways that can not be supplied by low-dimensional screens.
The distinction between low- and highdimensional phenotypes may sound technical, but it is crucial for choosing potential analysis methods. The central difference is that high-dimensional phenotypes

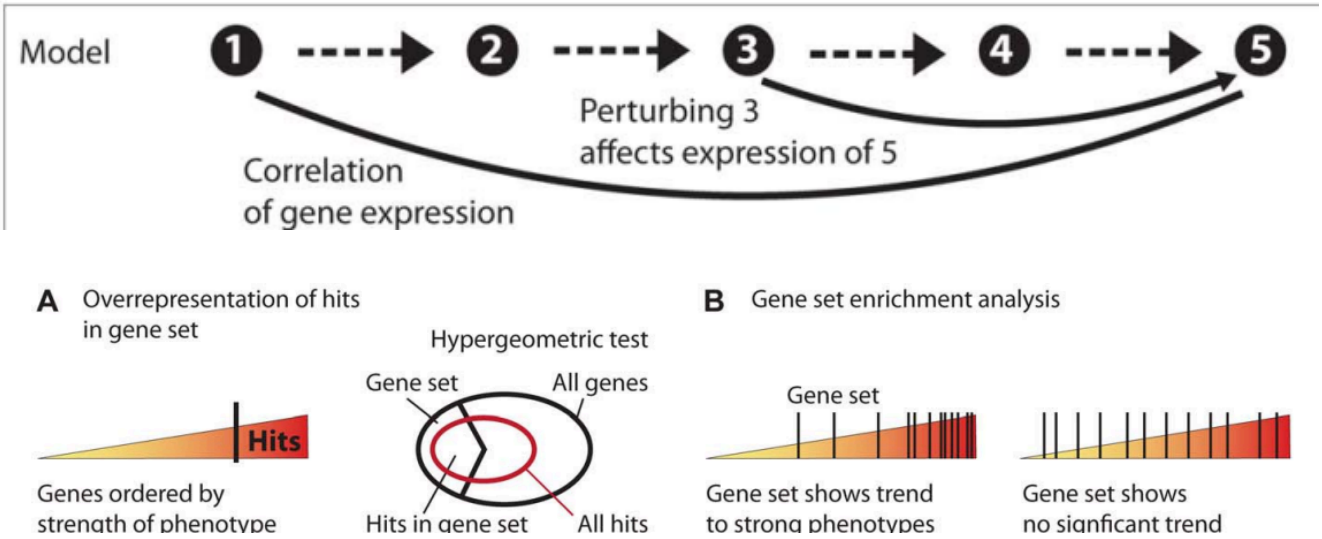
allow one to compute correlations and other similarity measures, which are not applicable for low-dimensional phenotypes.
plicable for low-dimensional phenotypes. Another important distinction is between static phenotypes, providing a ‘‘snapshot’’ of a cell’s reaction to a gene perturbation, and dynamic phenotypes showing a cell’s reaction over time

For the biological interpretation of screening results it is very important to keep in mind which level of ‘‘cellular granularity’’ a phenotype describes: growth rates or cell morphologies are much more ‘‘high-level’’ features of the cell than gene or protein expressions.

- Preprocessing Pipeline
  In this review we focus on single gene perturbations by knockouts [1] or RNAi [4] that allow targeting of individual genes or combinations of genes. Before network analysis, the raw data needs to pass an initial analysis and quality control pipeline specific to the perturbation and phenotyping technologies used
  Before network analysis, the raw data needs to pass an initial analysis and quality control pipeline specific to the perturbation and phenotyping technologies used. Low-dimensional screens are mostly performed in multiplewell-plates and a typical analysis pipeline [4] includes data preprocessing, removal of spatial biases per plate, normalization between plates, and finally detection of significant hits [6,7,24]. In vertebrates, genes need to be targeted with multiple siRNAs to ensure effective down-regulation [4], and the multiple phenotypes per gene can afterwards be integrated into a statistical score [25]. High-dimensional morphological screens depend on computational analysis like image segmentation [26,27] and phenotype discovery [28–30] for rapid and consistent phenotyping. Molecular high-dimensional phenotypes need preprocessing depending on their platform and different approaches exist, e.g., for flow-cytometry data [31] or microarrays.an indirect view on how different genes in the same process interact to achieve a particular phenotype.

  Gene expression phenotypes show transcriptional changes in the genes downstream of a perturbed pathway but offer only an indirect view of pathway structure because of the high number of nontranscriptional regulatory events like protein modifications
  A) Phenotypes are the response of the cell to external signals mediated by cellular networks and pathways. The goal of computation is to reconstruct these networks from the observed phenotypes. (B) Global molecular phenotypes like gene expression allow a view inside the cell but also have limitations.

  This gap between observed phenotypes and underlying cellular networks is the main problem in the analysis of perturbation screens and applies to both low- and highdimensional screens. The goal of computational analysis is to bridge this gap by inferring gene function and recovering pathways and mechanisms from observed phenotypes. The following methods address the challenge in different ways, mostly by integrating the perturbation effects and phenotypes with additional sources of information like collections of functionally related gene sets or protein-interaction networks
  A simple way to link phenotypes to gene function is to test whether pathways or functional groups of genes (e.g., defined by



(A) In the first approach [38] a cutoff is applied to select the hits with strongest phenotypes. A hyper-geometric test then evaluates if the overlap between the hits and a given gene set is surprisingly large (or small) compared to the overlap with a random set. (B) A second approach [35] does not need a cutoff. It maps the gene set (black bars) onto the observed phenotypes and quantifies if there is a significant trend or if the genes are spread out uniformly over the whole range.

Enrichment analysis results in a list of pvalues describing how significantly each gene set was represented in the hits. Enrichment analysis reduces complexity and improves interpretability of results by moving from single genes to functionally related gene sets. This type of analysis is often called ‘‘unbiased’’ and ‘‘hypothesis-free’’ and is ideal for a comprehensive first overview. However, enrichment analysis loses its value for complexity reduction if the number of gene sets becomes too big

per exploration of the data. For example, all enrichment methods rely on known gene sets and cannot uncover new pathways or components. Enrichment methods treat pathways as bags of unconnected genes without considering connections within and between pathways. Thus, enrichment methods can only deliver a very crude picture of the cell. In the following we will discuss approaches to overcome some of the limitations of enrichment analysis by integrating the observed phenotypes with complementary sources of information.

- Mapping Phenotypes to Networks
  An additional application of mapping hits to a network is that known phenotypes can be used to predict phenotypes of genes not included in the screen, e.g., by assuming that a gene connected to many hits should also show a strong phenotype [51]. The success of all network-mapping strategies strongly depends on the quality and coverage of both the screen and the linkage in the network.
- Gene Prioritization
  Gene prioritization [49,58] ranks genes according to how promising they would be for follow-up studies. Because it uses prior knowledge to fine-tune the algorithm, gene prioritization can be more focussed than a global uninformed search for enriched subnetworks.

<u>Network Analysis of HighDimensional Phenotypes</u>

- Global Overview by Clustering and Ranking
  ‘‘guilt-by-association’’ paradigm: genes with similar phenotypes will most probably have a similar biological function. This explains the prevalence of clustering techniques in analyzing highdimensional phenotyping screens

Clustering is a convenient first analysis and visualization step that can highlight strong trends and patterns in the data and can thus yield a global first impression of functional units. Another analysis strategy relying on guilt-by-association is to rank genes by their phenotypic similarity compared to a gene of interest [11]. Clustering and ranking can be combined with enrichment analysis (as discussed above) for functional interpretation.

- Graph Methods Linking Causes to Effects
  Another useful data visualization especially for transcriptional phenotypes is to build a directed (not necessarily acyclic) graph by drawing an arrow between two genes if perturbing one results in a significant expression change at the other [59]. This graph representation can be then used as a starting point for further analysis, for example by using graphtheoretic methods of transitive reduction [60] to distinguish between direct and indirect effects of a perturbation

- Probabilistic Graphical Models
  Most approaches to infer pathway structure from experimental data rely on probabilistic graphical models. For low-dimensional phenotypes they often suffer from nonuniqueness and unidentifiability issues , but can be applied very successfully in high-dimensional settings.
  A prominent approach are (static or dynamic) Bayesian networks, which describe probabilistically how a gene is controlled by its regulators [64,65]. To model experimental perturbations most approaches rely on the concept of ‘‘ideal interventions’’ [66], which deterministically fix a target gene to a particular state (e.g., ‘‘0’’ for a gene knockout). Ideal interventions were applied in Bayesian networks [21,67,68], factor graphs [69], and dependency networks [70]. In simulations [71,72] and on real data [21,71] it was found that interventions are critical for effective inference

  The model of ideal interventions contains a number of idealizations (hence the name), most importantly that manipulations only affect single genes and that perturbation strength can be controlled deterministically. The first assumption may not be true if there are off-target or compensatory effects involving other genes. The second assumption may also not hold true in realistic biological scenarios; in particular for RNAi screens where experimentalists often lack knowledge about the exact knock-down efficiency. Probabilistic generalizations of ideal interventions can be used to cope with this uncertainty

- Probabilistic Data Integration
  High-dimensional phenotypic profiles can be mapped to given graphs and networks by finding subgraphs that are connected in the background network and at the same time show high similarity of phenotypic profiles. These approaches already exist for mapping gene expression data onto protein interaction networks [74] and the same algorithms could easily be applied to any other kind of high-dimensional phenotypic profiles (see Figure 3B). Other approaches use data integration to construct potential pathways from protein interactions and transcription factor binding data to relate perturbed genes to the observed downstream effects

- Multiple Input - Multiple Output (MIMO) Models
  Many of the approaches discussed so far—like clustering or graphical models— can be applied to both static ‘‘snapshots’’ as well as dynamic time-course measurements. Another approach to model specifically the dynamics of networks comes from a branch of control theory called ‘‘systems identification’’ [78] and uses so called Multiple Input - Multiple Output (MIMO) models. MIMO models represent the evolution of a perturbed cell over time by linear

differential equations [79–83] and can represent nonlinear effects by transfer functions [84]. The models can be inferred by regression techniques in the linear case [80] or Monte Carlo stochastic search in the nonlinear case [84]. The framework is very flexible and can incorporate single as well as combinatorial perturbations.

- Nested Effects Models (NEMs)
  One of the key problems in analyzing perturbation screens is that the observed phenotypes are downstream of the perturbed pathway and may not show the direct influence of one pathway component on another. A class of models explicitly addressing this problem are Nested Effects Models (NEMS) [33,85]. They reconstruct pathway structure from subset relations on the basis of the following rationale: Perturbing some genes may have an influence on a global process, while perturbing others affects subprocesses of it. Imagine, for example, a signaling pathway activating several transcription factors.
  Blocking the entire pathway will most probably affect all targets of all transcription factors, while perturbing a single transcription factor will only affect its direct targets, which are a subset of the phenotype obtained by blocking the complete pathway. Given high-dimensional phenotypes showing a subset structure, NEMs find the most likely pathway topology explaining the data. They differ from other statistical approaches like Bayesian networks by encoding subset relations instead of correlations or other similarity measures.
  The theory of NEMs has been applied and extended in several studies [86–89]. An implementation is available as an R/Bioconductor package [90]. Other extensions to the NEM framework distinguish between activating and inhibiting regulation [91] or include dynamic information from time-series measurements [92].

In this review we have discussed two main approaches to describe the reaction of a cell to an experimental gene perturbation: low-dimensional phenotypes measure individual reporters for cell viability or pathway activation, while high-dimensional phenotypes show global effects on cell morphology, transcriptome, or proteome.

Instead of predicting networks from phenotypes, these approaches predict phenotypes from networks

From Single to Combinatorial Perturbations:
While single gene perturbation screens have been immensely successful in extending our knowledge of pathway components and interactions, an important limitation can be caused by compensatory effects, genetic buffering, and redundancy of cellular mechanisms and pathways [105,106]. This limitation can only be overcome by perturbing several genes at the same time. The number of possible combinations grows rapidly and thus current approaches are mainly limited to perturbing pairs of genes and observing low-dimensional phenotypes like fitness estimates [107]. The analysis of combinatorial perturbations is outside the scope of this review.